Edward Denton

+44 7865 618205 | EdwardDenton24@outlook.com | Edward Denton | RexMortem

Education

University of Warwick 2023-2026

Computer Science BSc

- Current Grade: **First**
- Relevant Modules: Functional Programming (**Haskell**), Data Structures (**Java**), Computer Organisation and Architecture (**C**), Algorithmic Graph Theory, Probability and Statistics, Groups and Rings, Databases

King's School Grantham 2016-2023

- **A-A*** in A-Levels (Maths, Further Maths, Physics, and Computer Science)
- **Computer Science Club**: Founded a club to support the extra-curricular development of students

Experience

Arm June 2025 - September 2025

SWE Intern (Fast Models)

- Implemented features of a CPU component generation script for CPU simulation, including generating test cases
- Migrated tests from one framework to another, automated using a Python script, to improve development workflow

University of Warwick Computing Society March 2024 - Present

Academic Officer

- Used pedagogical principles in producing resources for a 4-session **C Course** aimed at first year CS students
- Lead a 6-week collaboration with CodeSoc to run the **XSoc LeetCode Workshops** to aid technical interview prep
- Produced and presented **revision lectures** (e.g. for algorithms and automata) for exams and class tests

Mystifine Productions (Game Development) June 2020 – November 2020

Programmer and Administrator

- Developed multiple systems including a robust back-end database system in **Lua** and ROBLOX's datastore API, ensuring effective data persistence for player and guild data
- Applied best practices (DRY, YAGNI, Design Patterns) to enhance code readability and adaptability
- Managed a community of over 30000 members, addressing player feedback for continuous improvement

Projects

WildFire Modeller and Predictor (ICHack Win) | *Pixi, Flask, React, Matplotlib, Python, js, HTML, CSS*

- Worked in a 6-person team to produce a stochastic **percolation-based simulation** of wildfires in California
- Implemented Monte-Carlo simulations to produce a probability map for evaluating fire mitigation strategies
- Additionally applied TFT models to forecast wildfire spread overtime, using real-world datasets
- Produced in **24 hours for ICHack'26** and **won HRT's prize** for "Best use of Data for Predictions & Decision Making"

CSNotes Project (Project Lead) | *Flask, Shadcn (and React), UV, Docker, KaTeX, Gemini (LLM)*

- Integrated LLMs (Gemini) to generate AI summaries of module reviews, with caching to avoid redundant API calls
- Designed the UI using Shadcn and Tailwind, including display themes (e.g. high-contrast) saved in local storage
- Built interactive quizzes with matching, multi-answer, and drag-to-sort questions, plus markdown/LaTeX rendering

Chess and AI | *Python, Pygame, timeit*

- Benchmarked using timeit to identify bottlenecks in AI runtime to aid optimising the program
- Created custom event-driven architecture for UI using OOP and Pygame in Python
- Implemented rules of chess, and also an AI with the min-max algorithm and alpha-beta pruning

Logo Parser and Simulator | *Haskell, Gloss*

- Used Megaparsec to parse Logo code into an Intermediate Representation
- Interpreted, with the State monad, Logo instructions into an ADT representing state
- Simulated the Logo animation with Gloss from ADTs
- Implemented a CLI as an entry-point, complete with parsing error messages, to visualise simulations

Other projects available at https://github.com/RexMortem

Technical Skills

Languages: Lua, C, Haskell, Rust, Python, C++, Java, JavaScript, HTML, CSS
Tooling: Git, GitHub, Gerrit, Docker, Portainer, Figma, Jira
Frameworks/Tech: Flask, jQuery, Gloss, node.js, PyGame, Raylib

Interests

Climbing (Bouldering), Calisthenics, EdTech, Game Development, Competitive Programming